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EEE 202-1

LAB-04: Frequency Response of RL and RC Circuits

Introduction:

The goal of this assignment is to design and analyze an assigned filter type and values using transfer functions. For the first part, component values will be chosen according to the filter type and assigned corner frequency values in order to observe the required plot given on the Lab 4 document. For the second part, an RC circuit will be added onto our first design to observe the second part of the assignment document. The values for the added parts will also be determined in accordance to the assigned corner frequencies. For both parts, an analysis will be conducted to derive the formula for the transfer function for both cases. The transfer function behavior will be derived and plotted using MATLAB, and the circuits will be implemented in LTSpice to compare both results. After these steps, the circuits will be realized in real-life and their behavior will be compared to the simulations.

Analysis:

Before any derivations were to be made, the assignment sheet was checked on Moodle to determine the type of design we were given. It was observed that the C-type filter was assigned (Figure A) and the corner frequencies were given to be 500 kHz, 8 MHz and 50 kHz for f_1 , f_2 and f_3 respectively. This would mean that the point between the start and f_1 would have to be approximately -24 dB, as there are exactly four octaves between the corner frequencies for this assignment.

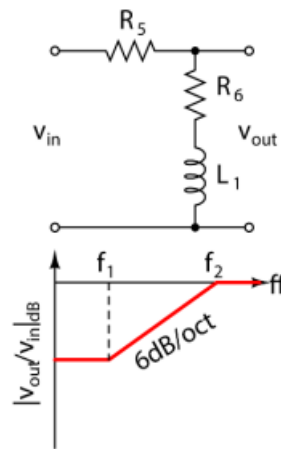


Figure A: Schematic of the C-type Filter

From this fact, and the requirement of the inductor being $15 \mu H$, the filter could be designed. It should be noted that since the inductor has a high quality-factor, its series resistance could be ignored practically, and the analysis will be conducted in accordance to this assumption. First, we know that we should observe approximately -3 dB at the corner frequency f_2 . For this to happen, the ratio between the output voltage and the input voltage should yield a value of $\sqrt{\frac{1}{2}}$ multiplied by the maximum value. This is achieved at infinite frequency and the maximum ratio of the voltages is 1. By utilizing the impedance formula for an ideal inductor of $Z_L = j\omega L$, we can use a simple voltage divider in the phasor domain to derive an equation. Since $\omega = 2\pi f$ and $f_2 = 8 \text{ MHz}$, we can say that our inductor will have an impedance of $j753.9822 \text{ Ohms}$ at f_2 . Therefore, we have the relation of:

$$\left| \frac{V_{out}}{V_{in}} \right| = \frac{\sqrt{R_6^2 + 753.9822^2}}{\sqrt{(R_5 + R_6)^2 + 753.9822^2}} = \sqrt{\frac{1}{2}} \quad (1)$$

At the corner frequency f_2 . Another factor that would have to be considered is the ratio between the resistors. Since we have a net gain of -24 dB at frequencies f_1 , and since the impedance of the inductor is negligible at low frequencies -which means that it could be treated as a short circuit-, we can determine the ratio of the resistors. Since the formula for gain in decibels is determined by $20 \log\left(\left| \frac{V_{out}}{V_{in}} \right| \right)$, we can find that the ratio between the output voltage and input voltage is equal to $10^{-1.2}$ by rearranging and equation the relation to -24. This value is approximately 0.0631, which means that the ratio between R_6 and net resistance is 0.0631. Likewise, the ratio of the net resistance to R_6 would be the inverse, which is approximately 15.8489. By and rearranging equation (1) whilst using this newfound relation, we can write:

$$\frac{R_6^2 + 753.9822^2}{(15.8489R_6)^2 + 753.9822^2} \approx \frac{1}{2} \rightarrow R_6 = \pm \sqrt{\frac{753.9822^2}{15.8489^2 - 2}} \quad (2)$$

Since resistance cannot be negative, we take the positive solution for R_6 , which is 47.7637 Ohms. Since we will need to construct this circuit in real life, we will use 47 as the ohmic value for the resistor when required. This would yield a small error of 1.5989%, which is negligibly small. And since we know the ratio of the net resistance to R_6 , which is 15.8489, we can say that the net resistance has to be approximately 757.0021. By subtracting the value of R_6 , we know that R_5 has to be approximately 709.2384 Ohms. Since this is not a standard value, we can use a 680 Ohm resistor and a 27 Ohm resistor in series to approximate R_5 when the circuit will be constructed in real-life. This will yield a component error of 0.3156%, which is negligible. Now that the values for the resistors have been determined, further preliminary work can be done.

Preliminary Part 1/2 (Filter Design and Plot):

As is mention in the Analysis section, we can treat the system like a voltage divider in the phasor domain. If component values are not inserted, the transfer function can be expressed as:

$$H(j\omega) = \frac{V_{out}}{V_{in}} = \frac{R_6 + j\omega L}{R_6 + R_5 + j\omega L} \rightarrow |H(j\omega)| = \frac{\sqrt{R_6^2 + \omega^2 L^2}}{\sqrt{(R_5 + R_6)^2 + \omega^2 L^2}} \quad (3)$$

And if the derived component values are inserted into equation (3), we have:

$$|H(j\omega)| = \frac{\sqrt{(47.7637)^2 + (15 \cdot 10^{-6})^2 \omega^2}}{\sqrt{(757.0021)^2 + (15 \cdot 10^{-6})^2 \omega^2}} \quad (4)$$

Which is the transfer function for this case. After this relation was derived, the function in equation (4) was plotted using MATLAB (Figure B).

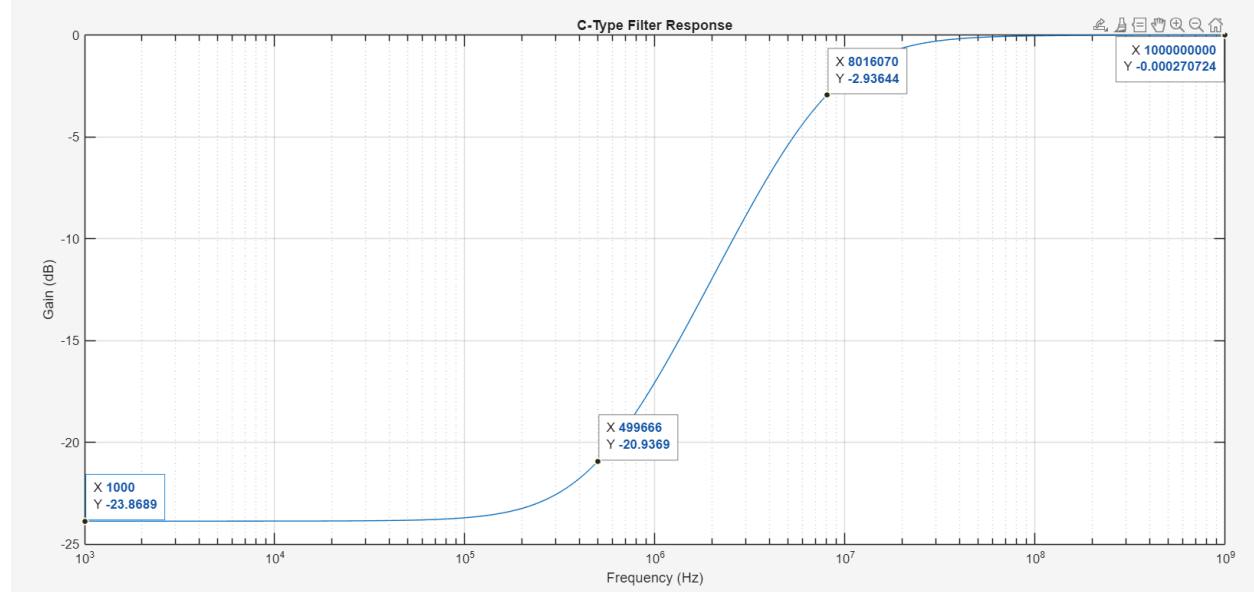


Figure B: MATLAB Plot of the C-type Filter (Code in Appendix)

After the plot was observed, the gain at the corner frequencies and at the endpoints were marked. Since the corner frequencies are to have 3 dB deltas to the high and low frequency asymptotes, and since the high and low frequency asymptotes were 0 dB and -24 dB for our specific case, we expected a gain of -21 dB and -3 dB at f_1 and f_2 respectively. An error of 0.5463%, 0.3005% and 2.1190% were observed for the low frequency response, f_1 and f_2 respectively. These could be caused by rounded errors made in the calculation and floating-point errors. It should be noted that since we expected 0 dB at the high frequency response, we could not calculate error at that point as it would require a division by zero, which is not defined. However, the gain at the high frequency was incredibly low, so this task was deemed a success.

Preliminary Part 3 (Corner Frequency Derivation):

In order to compute the corner frequencies, we can set the ωL terms in the transfer function to the constants' values in the numerator and denominator respectively. The corner frequencies are found by equating the constant and frequency-dependent terms inside each square root, and since we are asked to use standard component values, a 47 Ohm resistor's value will be used for one corner frequency, and 754 Ohms -which is the sum of 47, 680 and 27 Ohm resistors totaling the net resistance of the circuit (see Analysis)- for the other. Therefore, we have the relation:

$$\omega_1 = \frac{R_6}{L}, \quad \omega_2 = \frac{R_5 + R_6}{L} \quad (5)$$

Where L is $15 \mu H$, R_6 is 47 Ohm and R_5 is 707 Ohm. Using these values and the definition $\omega = 2\pi f$, we see that the derived corner frequencies f_1 and f_2 are 498.6855 kHz and 8000.1885 kHz respectively. Recalling the assigned values of 500 kHz and 8000 kHz from the Lab 4 Assignment sheet, we can find

our error to be 0.2629% and 0.0024% respectively for frequencies f_1 and f_2 . This small error can be caused by not choosing precise enough values for the resistors.

Preliminary Part 4 (LTSpice Simulation):

After the component values and corner frequencies were justified for the real-life implementation, the circuit as constructed using LTSpice with AC Analysis parameters. The response of the circuit was cursoried at multiple points, such as the responses at f_1 and f_2 (Figure C), low frequencies, high frequencies (Figure D) and the roll-off per octave (Figure E).

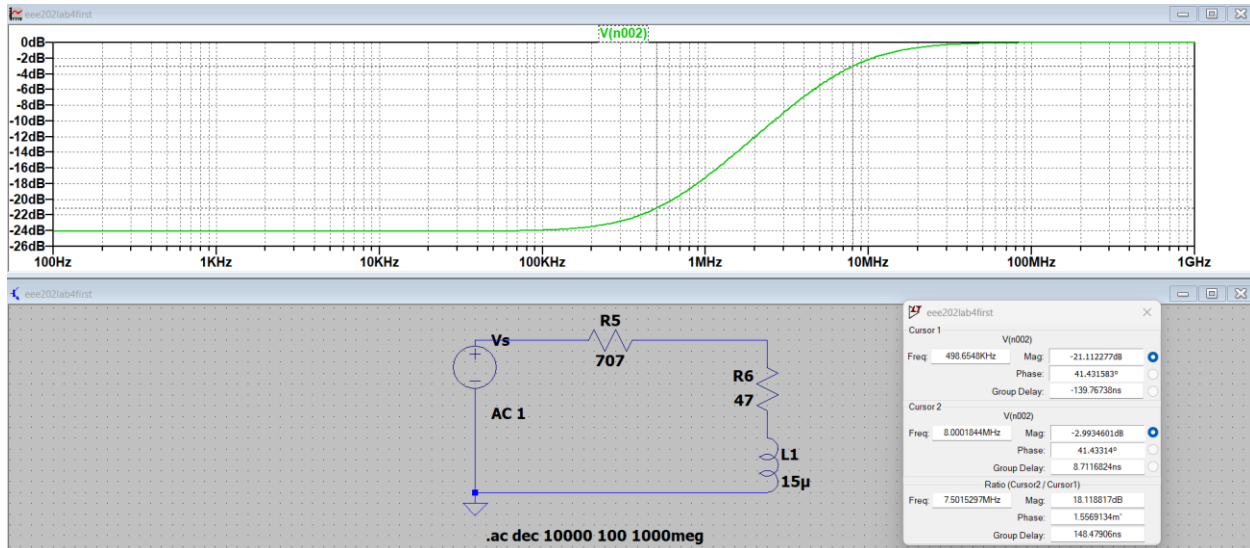


Figure C: LTSpice Simulation of the C-type Filter with corner frequency responses cursoried

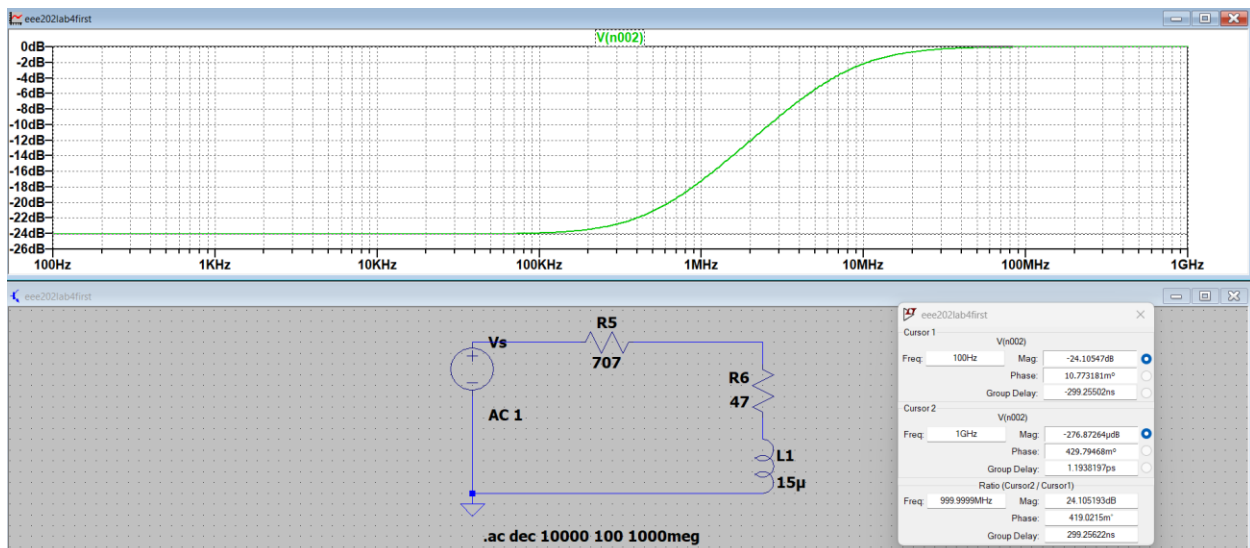


Figure D: LTSpice Simulation of the C-type Filter with high and low frequency responses cursoried

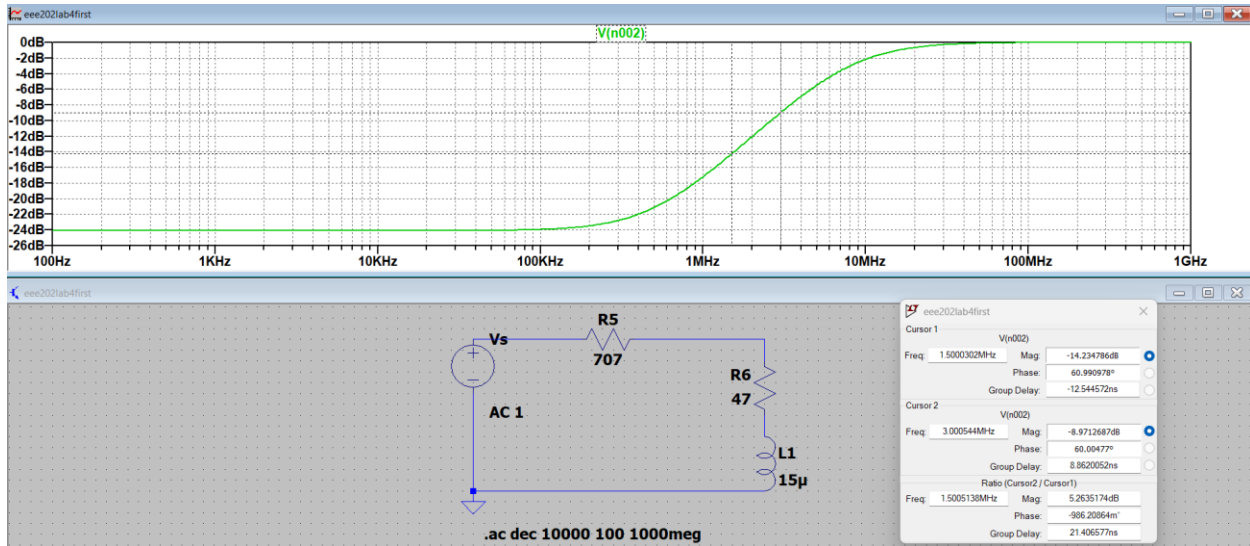


Figure E: LTSpice Simulation of the C-type Filter with frequency responses for roll-off slope cursored

By inspection, the overall shape of the graph matched the given plot on the Lab 4 assignment sheet. We expected the response at f_1 to be 3 dB higher than the low frequency response, and a difference of 2.9932 dB was observed. Likewise, the response at f_2 was expected to be 3 dB lower than the high frequency response, and a difference of 2.9932 dB was observed. This meant that an error of 0.2267% was observed for both the corner frequencies. This could be caused by approximate component values and floating-point errors of the software.

Additionally, the roll-off octave between the corner frequencies were measured to be 5.2623 dB/octave whereas we expected 6 dB/octave for the ideal case. This meant that an error of 12.2950% was observed for this case. This can be caused by the corner frequencies being too close to each other and influencing the range between them. A method to reduce the error would be to space out the corner frequencies more, however, since they have been assigned, they cannot be changed. Due to this fact, the task was deemed to be a success.

Preliminary Part 5/6 (Third Corner Frequency Addition and Transfer Function):

In order to add a third corner frequency to our circuit, we were tasked to append an RC circuit to our previous circuit. The frequencies less than the third corner frequency (which is assigned to be 50 kHz) will have a slope of 6 dB/octave (Figure F).

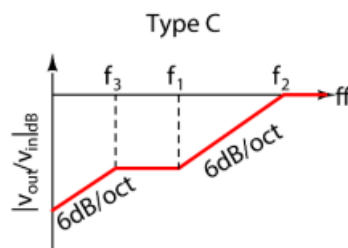


Figure F: Ideal response of the updated C-type circuit

Since, for this plot, we expect almost zero voltage on the impeding block with voltage V_{out} . In order to do this, we can have a high impedance block behind our measured node for low frequencies. This can be

done by adding a series capacitor and resistor between V_{in} and R_5 . Letting the value for these arbitrary capacitor and resistor be C and R respectively, and utilizing the fact that, in the phasor domain, the impedance of the capacitor is equated to be $Z_C = \frac{1}{j\omega C}$, the new transfer function of our design will be:

$$H(j\omega) = \frac{R_6 + j\omega L}{(R_5 + R_6 + R) + j(\omega L - \frac{1}{\omega C})} = \frac{47\Omega + j\omega(15\mu H)}{(754\Omega + R) + j(\omega(15\mu\Omega) - \frac{1}{\omega C})} \quad (6)$$

After the component values that have been derived for our real-life implementation. Taking the magnitude of this equation, we get:

$$|H(j\omega)| = \frac{\sqrt{47^2 + (15 \cdot 10^{-6})^2 \omega^2}}{\sqrt{(754 + R)^2 + \left((15 \cdot 10^{-6})\omega - \frac{1}{\omega C}\right)^2}} \quad (7)$$

For future derivations, the inductance will be denoted by L for readability. Since corner frequencies are found when they're equal to their respective constants for the denominator and numerator, we can equate the frequency dependent terms to their constants. First, we can apply this process to the numerator to find that:

$$\omega_{c1} = \frac{47\Omega}{L} \rightarrow f_{c1} = 498.6855 \text{ kHz} \quad (8)$$

Which is equal to the corner frequency of the previous circuit. Now, we can rearrange the denominator to get:

$$\left|\omega L - \frac{1}{\omega C}\right| = |754\Omega + R| \rightarrow L\omega^2 - (754\Omega + R)\omega - \frac{1}{C} = 0 \quad (9)$$

We can apply the quadratic formula to find the roots in equation (9). Keeping in mind that C and R have to be negative in the real world, we get:

$$\omega_{2,3} = \frac{754\Omega + R \pm \sqrt{(754\Omega + R)^2 + \frac{4L}{C}}}{2L} \quad (10)$$

Since we have two equations and two unknowns, this equation can be solved. However, by inspection, we can see that we will get a negative result when the negative part is considered, as the square root expression is larger than $(754+R)$. But since frequency cannot be negative, we can take the absolute value for that expression to get the actual frequency. Since f_2 and f_3 are 8 MHz and 50 kHz respectively, we can substitute in $R=0$ and consider the $\frac{4L}{C}$ to be much smaller than 754^2 Ohms to get f_2 , as we derived that value in Preliminary Part 3. Since we selected our arbitrary resistance R to be zero, we can find the capacitance by equating equation (10) to ω_3 . So, we can rewrite the equation as:

$$\omega_3 = \frac{754 - \sqrt{754^2 + \frac{4L}{C}}}{2L} \rightarrow \frac{4L}{C} = |(2\omega_3 L - 754)^2| - 754^2 \rightarrow C = \left| \frac{(2\omega_3 L - 754)^2 - 754^2}{4L} \right|^{-1} \quad (11)$$

If our value for L and ω_3 are inserted -which are $15\mu H$ and $100000\pi \text{ rad/s}$ respectively-, we can see that the capacitance is equal to 4.2481 nF. However, since this capacitance value doesn't exist as one package in real life, we can approximate it as 4.23 nF since this value can be achieved if a 3.9 nF and a

330 pF capacitor is placed in parallel. This will yield an error of 0.4261% for the value of our capacitor. If our derived values for C and R are inserted into equation (10), we get:

$$\omega_{2,3} = \frac{754\Omega \pm \sqrt{(754\Omega)^2 + (14123.6288)^2\Omega^2}}{30 \times 10^{-6}H} \quad (12)$$

Which yields 8.0498 MHz for f_2 and -49.5935 kHz for f_3 . As previously mentioned above, we can take the absolute value of our negative frequency to find the real value for it, which is 49.5935 kHz. If we compare these values to our ideal frequencies of 8 MHz and 50 kHz, we get an error of 0.6225% and 0.8130% respectively. These can be caused by us approximating the capacitor values and truncations during the derivations. It should be noted that our corner previous frequencies f_1 and f_2 have remained extremely close to our corner frequencies for our first design. Due to the small magnitude of the errors and the corner frequencies remaining the same, this part was deemed to be a success. Additionally, it should be noted that our transfer function, when our derived values are inserted, is equal to:

$$H(j\omega) = \frac{R_6 + j\omega L}{(R_5 + R_6) + j(\omega L - \frac{1}{\omega C})} = \frac{47\Omega + j\omega(15\mu H)}{754\Omega + j(\omega(15\mu H) - \frac{1}{\omega(4.23 \text{ nF})})} \quad (13)$$

Which fits our assigned response.

Preliminary Part 6 (LTSpice Simulation):

After the previous derivations were made for the component values and circuit structure, our designs were to be implemented in LTSpice. The described structure with the given component values were made and an AC analysis was performed using LTSpice. After the plot was generated, the gain at the new f_1 and f_2 (Figure G), high frequency response and gain at f_3 (Figure H), the slope between f_1 and f_2 (Figure I) and the slope before f_3 (Figure J) was analyzed.

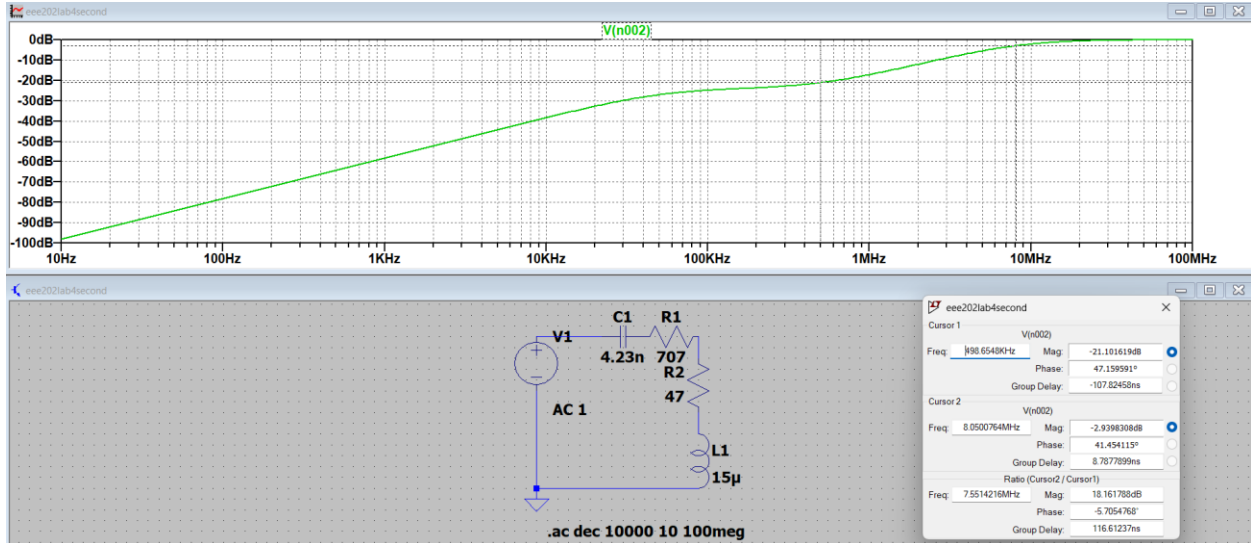


Figure G: Response of the updated C-type circuit at f_1 and f_2

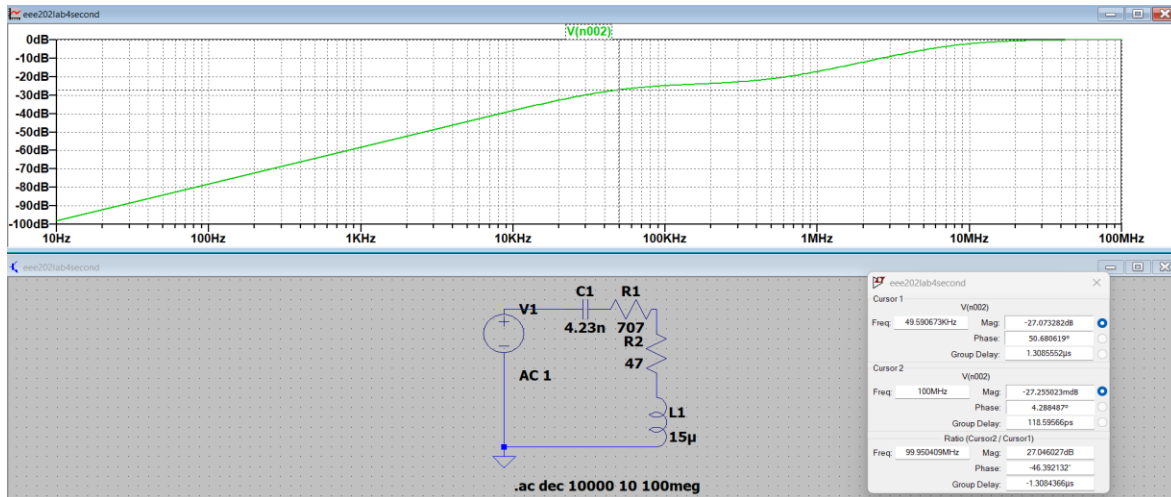


Figure H: Response of the updated C-type circuit at f_3 and high frequencies

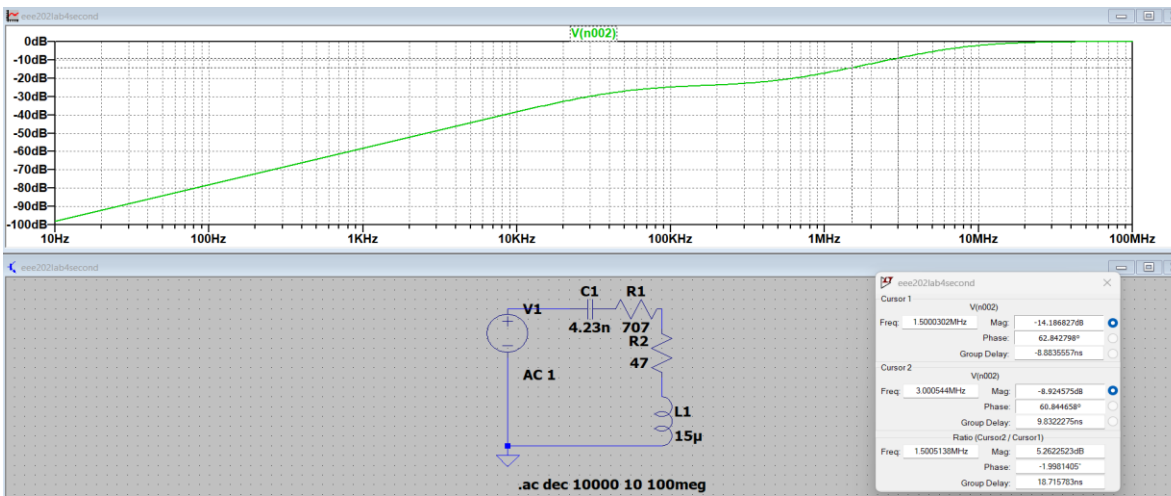


Figure I: Slope of the updated C-type circuit between f_1 and f_2

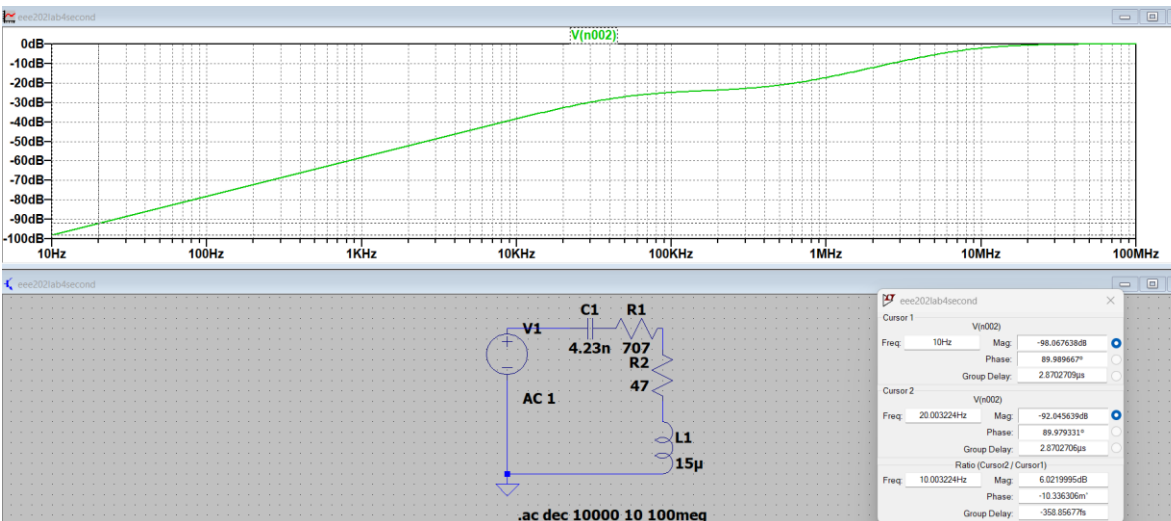


Figure J: Slope of the updated C-type circuit between low frequencies and f_3

The gain at the corner frequencies f_1 , f_2 and f_3 were found to be -21.1016 dB, -2.9398 dB and -27.0733 dB respectively. When compared with their ideal values of -21 dB, -3 dB and -27 dB respectively (3 dB deltas to their respective plateaus), a respective error of 0.4838%, 2.0067% and 0.02715% for each frequency. This could be caused by non-exact component values and human calculation errors. Additionally, the high frequency response was extremely close to but not exactly 0 dB, which could be explained by slight resistive losses present during all frequencies in the circuit.

Furthermore, the slope between f_1 and f_2 was determined to be 5.2610 dB/octave by the formula of $\frac{dB_2 - dB_1}{\log_2(\frac{f_2}{f_1})}$. When this was compared to our previous derived slope, which was 5.2623 dB/octave, an error of 0.0247% was observed. Since this error is extremely small, it was deemed that the behavior of the circuit above f_3 had not changed in a considerable manner, which was the desired behavior. The same was done for the slope behind f_3 , and it was determined to be 6.0206 dB/octave. Since our desired slope was 6 dB/octave, an error of 0.3433% was observed. This could be due to non-exact component values and truncation errors made by LTSpice.

Hardware Implementation:

After the preliminary steps were taken, the circuit was to be built in real-life. For the first design, two resistors of value 680 Ohm and 27 Ohm were soldered together to create a resistor of 707 Ohm due to the requirement of our design. A 47 Ohm resistor was placed in series with the soldered resistor, and a high Q 15 microHenry inductor was placed in series with it. Two power supply leads were connected to the circuit such that the input voltage was received to the free leg of the 707 Ohm and the ground was connected to the free leg of the inductor. After the leads were connected, the amplitude was set to 5 Volts peak-to-peak, and an oscilloscope probe was connected to the node between the two resistors. Finally, a coaxial cable was used to connect the external triggering port to the signal generator to properly trigger the measurement. After these steps, the design was created in real-life (Figure 1.1).



Figure 1.1: Implemented and measured C-Type Circuit

Before doing any measurements with our circuit, our LT-Spice model was updated with the measured values of the components. Additionally, to handle high-frequency behaviors in the circuit, a small parallel capacitance was added to the inductor the model the parasitic parallel capacitance. (Figure 1.2). After these steps, the simulation was ran, and datapoints were taken for the circuit (Table 1) with pictures for important values (Figure 1.3-1.8). Also, the datapoints were plotted using MATLAB (Figure 1.9).

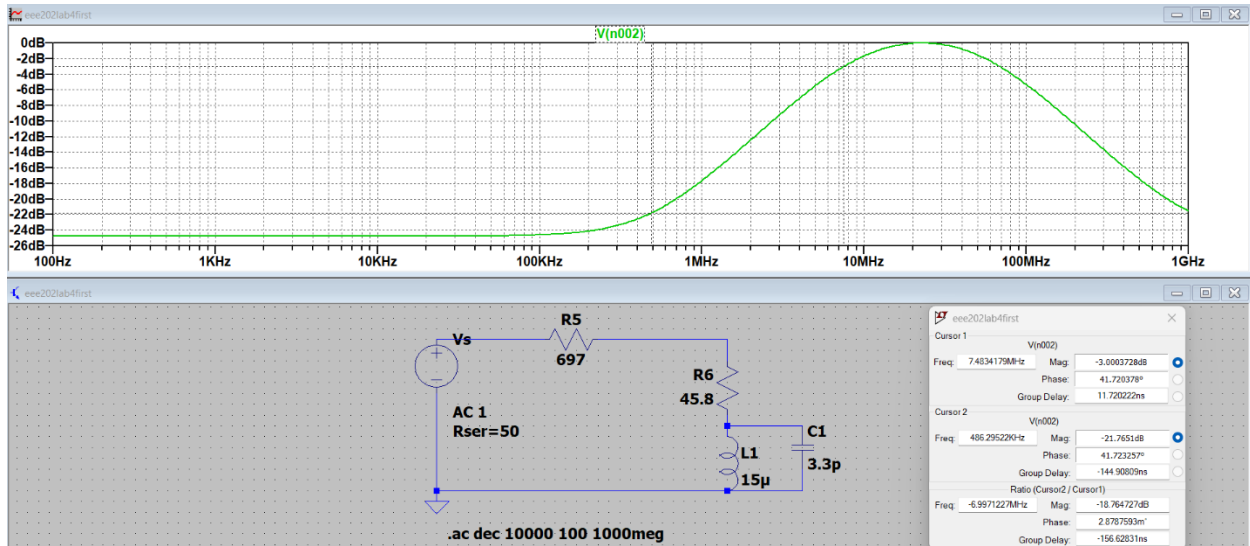


Figure 1.2: LTSpice simulation of the realistic C-Type Circuit

f	Freq (kHz)	V_{out}	V_{out}/V_{in}	V_{out}/V_{in} (dB)	f	Freq (kHz)	V_{out}	V_{out}/V_{in}	V_{out}/V_{in} (dB)
$0.1f_1$	49.86855	624 mV	0.0624	-24.110	$0.1f_2$	800.01885	1.80 V	0.18	-16.954
$0.2f_1$	99.7371	630 mV	0.0630	-24.013	$0.2f_2$	1600.0377	2.82 V	0.282	-10.995
$0.4f_1$	199.4742	664 mV	0.0664	-23.876	$0.4f_2$	3200.0754	5.20 V	0.52	-5.680
$0.8f_1$	398.9484	792 mV	0.0792	-22.025	$0.8f_2$	6400.1508	6.52 V	0.652	-3.715
f_1	498.6855	888 mV	0.0888	-21.037	f_2	8000.1885	7.44 V	0.744	-2.545
$1.2f_1$	598.4226	1.06 V	0.106	-19.494	$1.2f_2$	9600.2262	8.90 V	0.890	-1.012
$2f_1$	997.371	1.42 V	0.142	-14.895	$2f_2$	16000.377	7.42 V	0.742	-2.592

Table 1: Measured Data from the C-Type Circuit (Voltage in Peak-to-peak measurements)



Figure 1.3: Output Voltage of the C-Type Circuit at $0.1f_1$

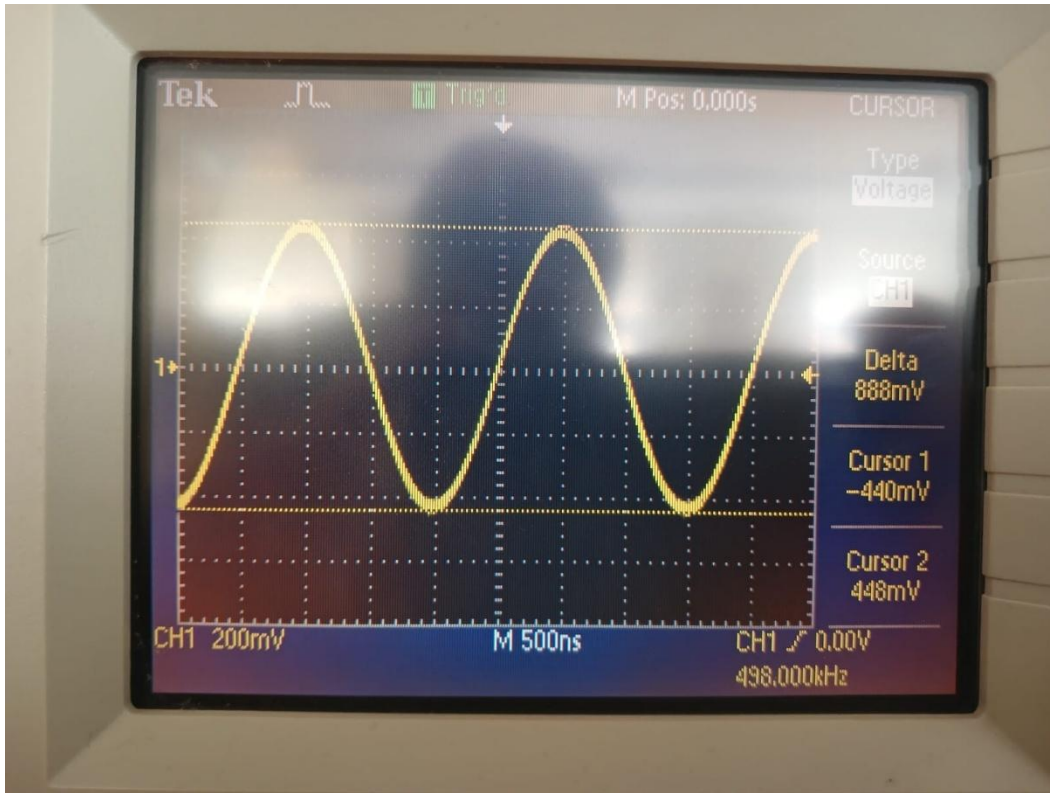


Figure 1.4: Output Voltage of the C-Type Circuit at f_1

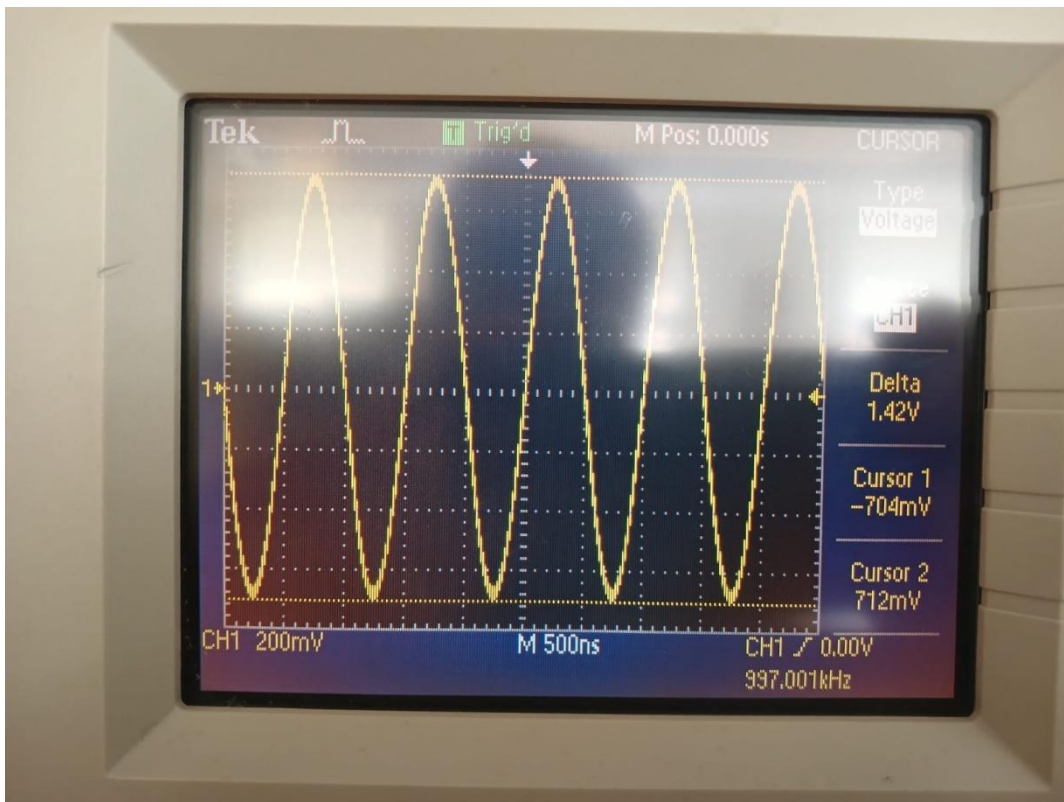


Figure 1.5: Output Voltage of the C-Type Circuit at $2f_1$



Figure 1.6: Output Voltage of the C-Type Circuit at $0.1f_2$

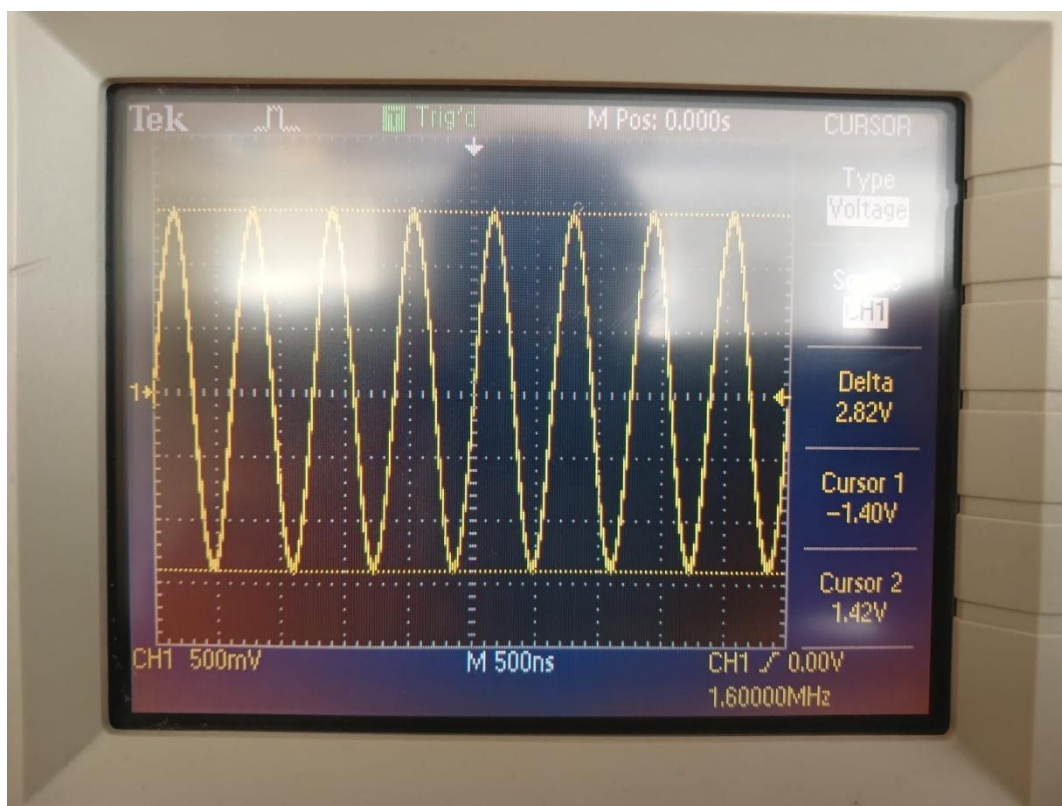


Figure 1.7: Output Voltage of the C-Type Circuit at $0.2f_2$

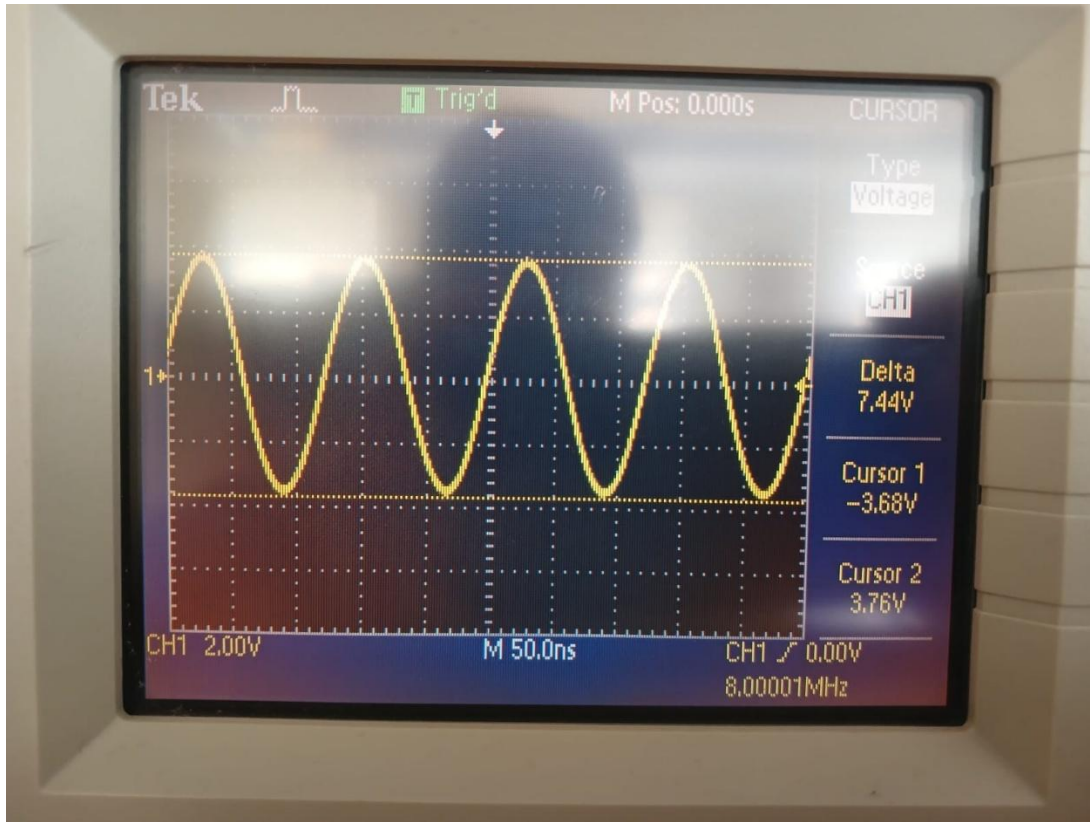


Figure 1.8: Output Voltage of the C-Type Circuit at f_2

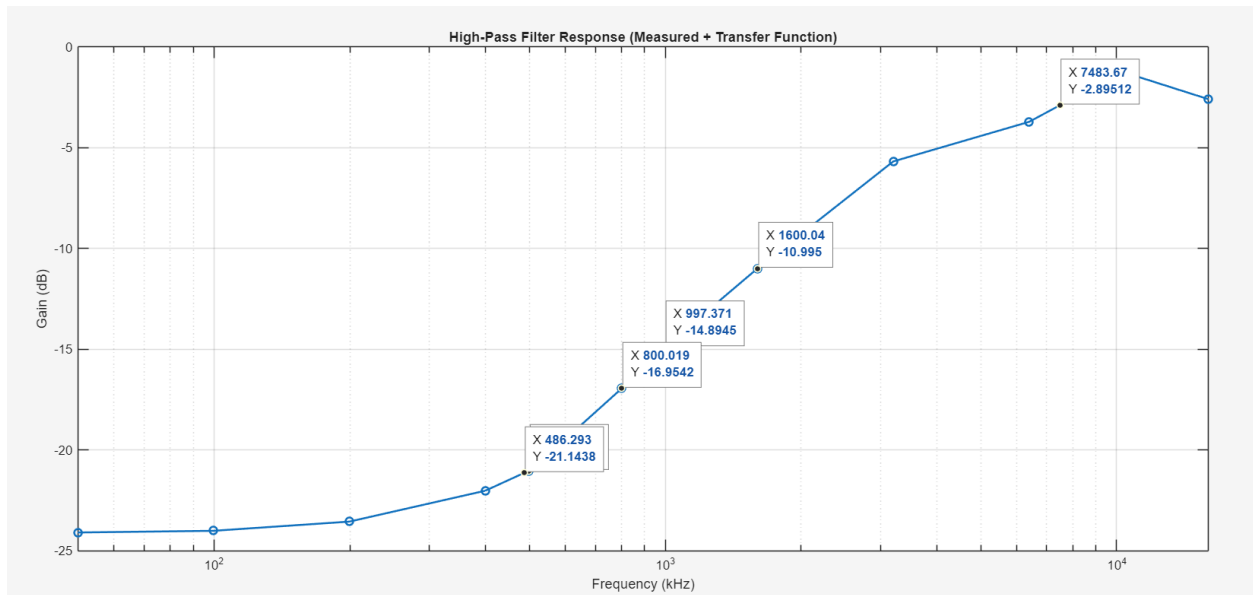


Figure 1.9: Semi-log plot of our data in Table 1 (Code in Appendix)

After these data entries were recorded, they were analyzed. From the updated LTSpice plot, the new corner frequencies were determined to be 486.2952 kHz and 7.4834 MHz for f_1 and f_2 respectively. From our original ideal derivations, these frequencies yield an error of 2.4846% and 6.4597% respectively for the corner frequencies. By analyzing the MATLAB plot, our corner frequencies had a gain of -21.1438 dB and -2.8951 dB for f_1 and f_2 respectively. When compared to the ideal values from our LTSpice plot of -21.7651 dB and -3.0003 dB, an error of 2.8546% and 3.5063% were gathered for f_1 and f_2 . This shift in frequencies and gain can be explained by our components' values not being fully exact and being approximated. Additionally, especially for high frequencies, our measuring methods tend to be more inaccurate as our probe is only rated for measurements up to 6 MHz.

On top of our corner frequencies, the slope of the graph was also analyzed. By looking at the frequency pairs of f_1 and $2f_1$; $0.1f_2$ and $0.2f_2$, and the respective gains between them, slopes of 6.142 dB/octave and 5.959 dB/octave were observed. Compared to the ideal slope of 6 dB/octave, respective errors of 2.367% and 0.6833% were gathered. Due to this small error -which has probable reason of non-exact component values and measuring errors- we have confirmed that our plot exhibits a ± 6 dB/octave in two regions. Additionally, comparing the real-life and LTSpice low-frequency gains, which were -24.110 and -24.7210 dB respectively, an error of 2.4716% was observed. This could be due to the sensitivity of the measurements for small voltage values. Finally, for the overall shape of the graph, it matched our LTSpice simulation's plot very closely until 6.4 MHz, where a drop in gain was seen. This could be due to the oscilloscope probe being rated for measurements until 6 MHz and the signal generator experiencing problems. It should be noted that a dip in gain was seen at the largest measured frequency of 16 MHz, which can be explained by the parasitic parallel capacitance of the inductor. After these measurements, the first part of the experiment was deemed as success and an RC circuit was to be added.

After doing the required tasks for the first part, our design in Figure G was to be created in real-life. The circuit from the first part was powered off and it's leads were taken off. Then the resistive branch was connected to a breadboard in series with two parallel capacitors whose rated values were 3.9 nF and 330 pF (Figure 2.1).

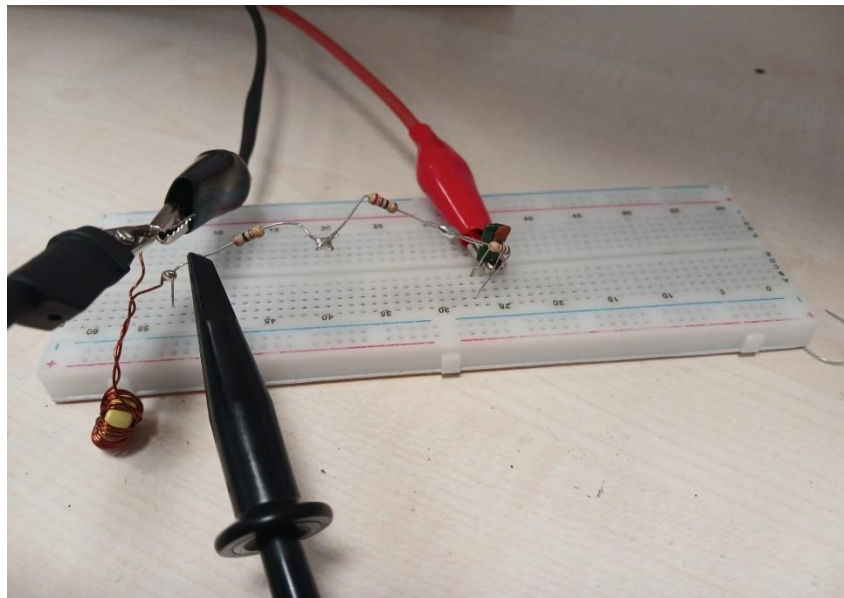


Figure 2.1: Implemented, measured and updated C-Type Circuit

The values of the components were measured and the LTSpice model was updated. As the same as part one of the assignment, a parasitic parallel capacitance was added to the inductor to model the high frequency behavior (Figure 2.2). After these steps, the simulation was ran, and datapoints were taken for the circuit (Table 2) with pictures for important values (Figure 2.3-2.9). Also, the datapoints were plotted using MATLAB (Figure 2.10).

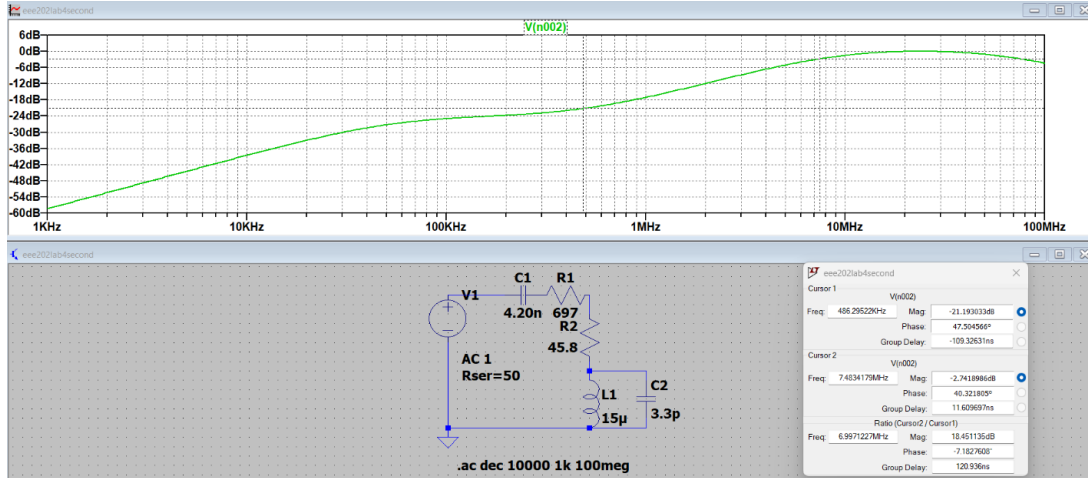


Figure 2.2: LTSpice simulation of the realistic modified C-Type Circuit

f	Freq (kHz)	V_{out}	V_{out}/V_{in}	V_{out}/V_{in} (dB)	f	Freq (kHz)	V_{out}	V_{out}/V_{in}	V_{out}/V_{in} (dB)
$0.1f_3$	5	78 mV	0.0078	-42.158	$1.5f_1$	750	1.06 V	0.106	-19.494
$0.2f_3$	10	158 mV	0.0158	-36.027	$2f_1$	1000	1.23 V	0.123	-18.202
$0.5f_3$	25	308 mV	0.0308	-30.229	$5f_1$	2500	3.00 V	0.300	-10.487
f_3	50	428 mV	0.0428	-27.371	f_2	8000	7.52 V	0.752	-2.476
$0.2f_1$	100	518 mV	0.0518	-25.713	$2f_2$	16000	9.60 V	0.960	-0.355
$0.5f_1$	250	643 mV	0.0643	-23.836	f_x	22000	6.68 V	0.668	-3.505
f_1	500	880 mV	0.0880	-21.110	f_{max}	30000	3.72 V	0.372	-8.589

Table 2: Measured Data from the modified C-Type Circuit (Voltage in Peak-to-peak measurements)

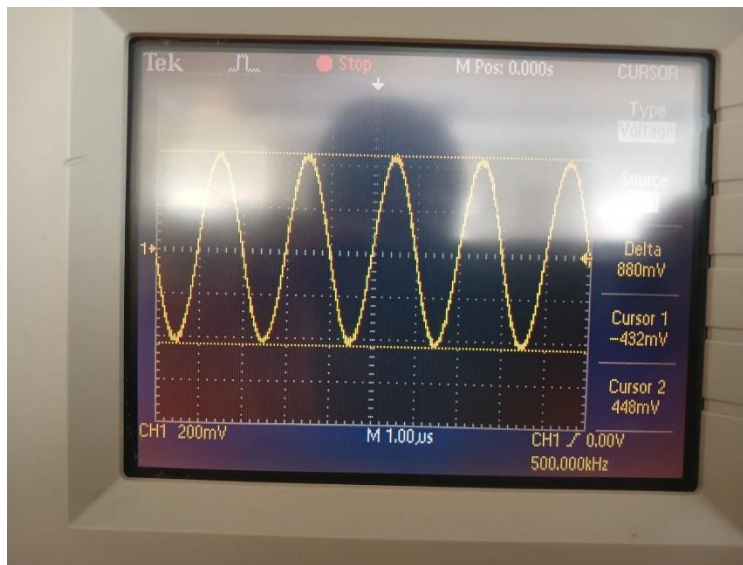


Figure 2.3: Output Voltage of the modified C-Type Circuit at f_1

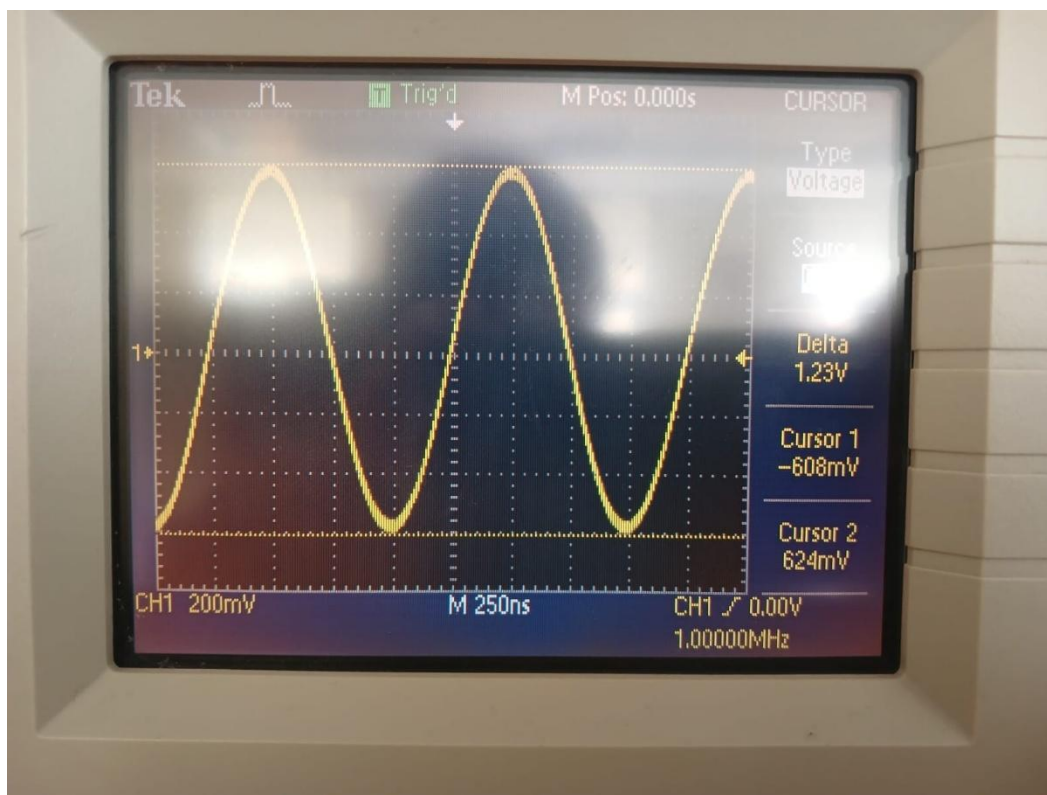


Figure 2.4: Output Voltage of the modified C-Type Circuit at $2f_1$

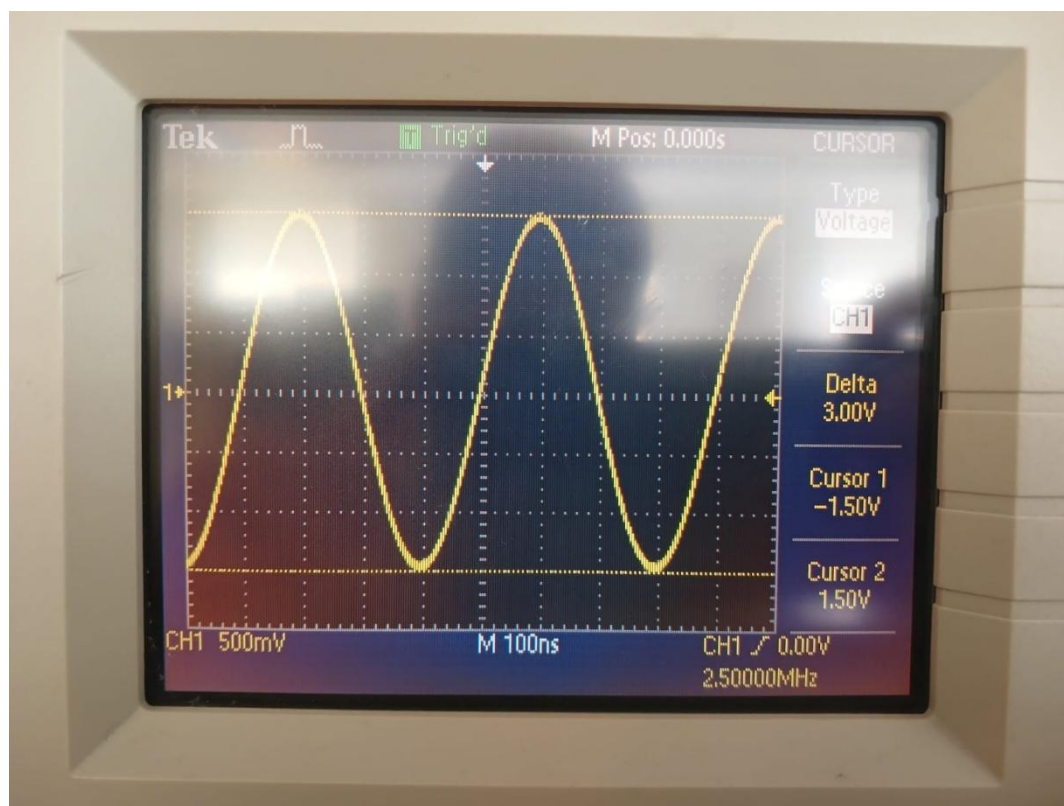


Figure 2.5: Output Voltage of the modified C-Type Circuit at $5f_1$

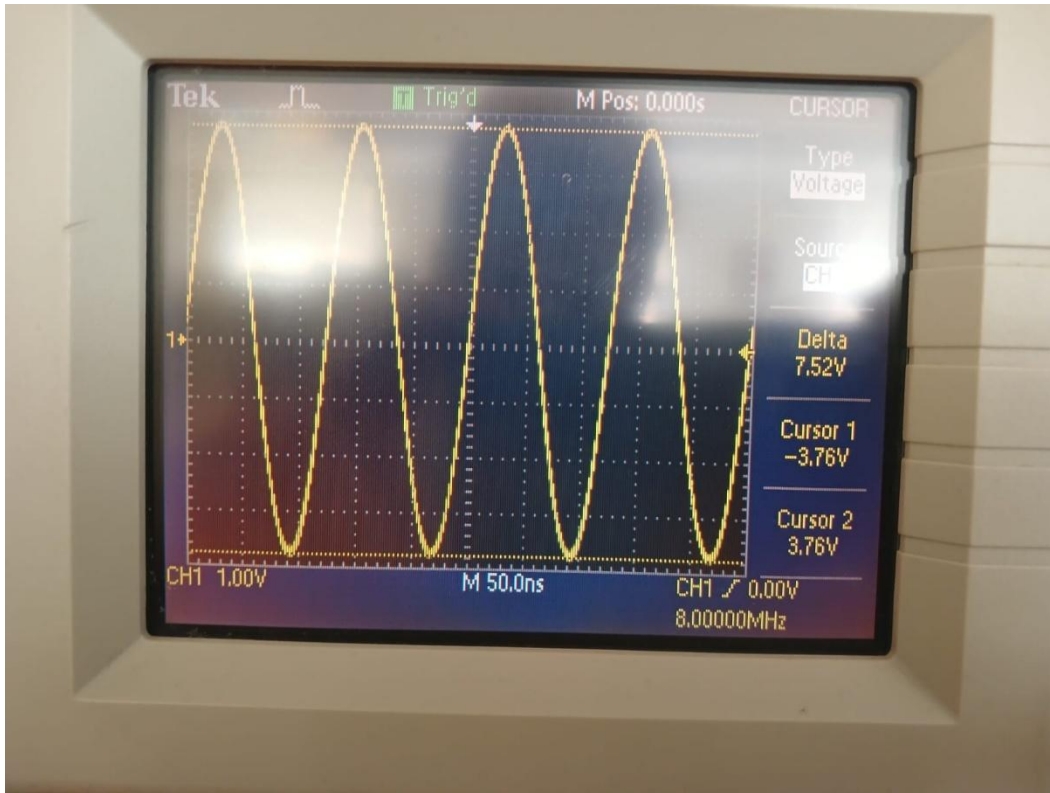


Figure 2.6: Output Voltage of the modified C-Type Circuit at f_2

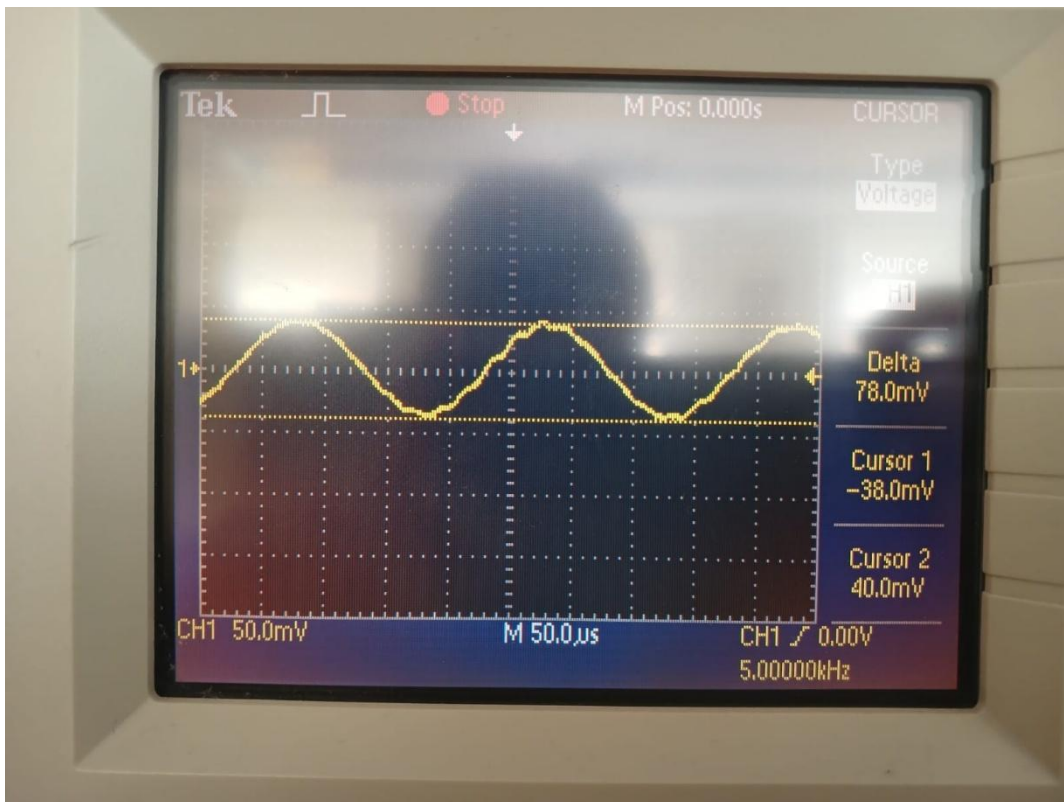


Figure 2.7: Output Voltage of the modified C-Type Circuit at $0.1f_3$

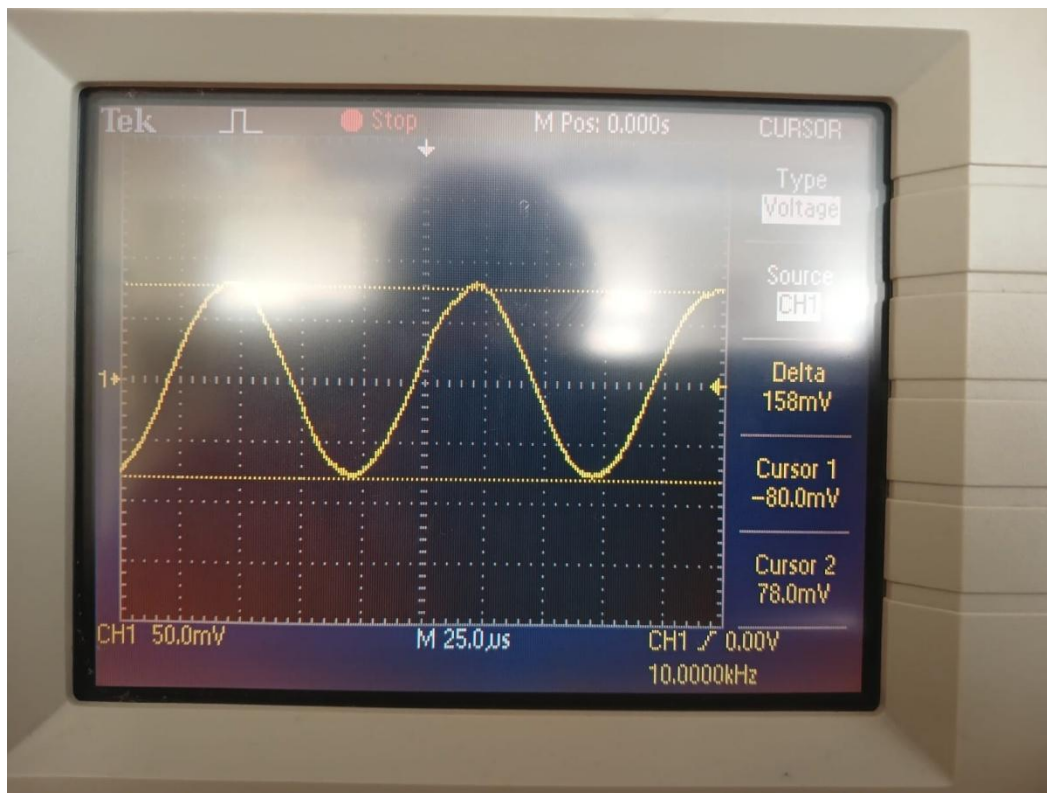


Figure 2.8: Output Voltage of the modified C-Type Circuit at $0.2f_3$

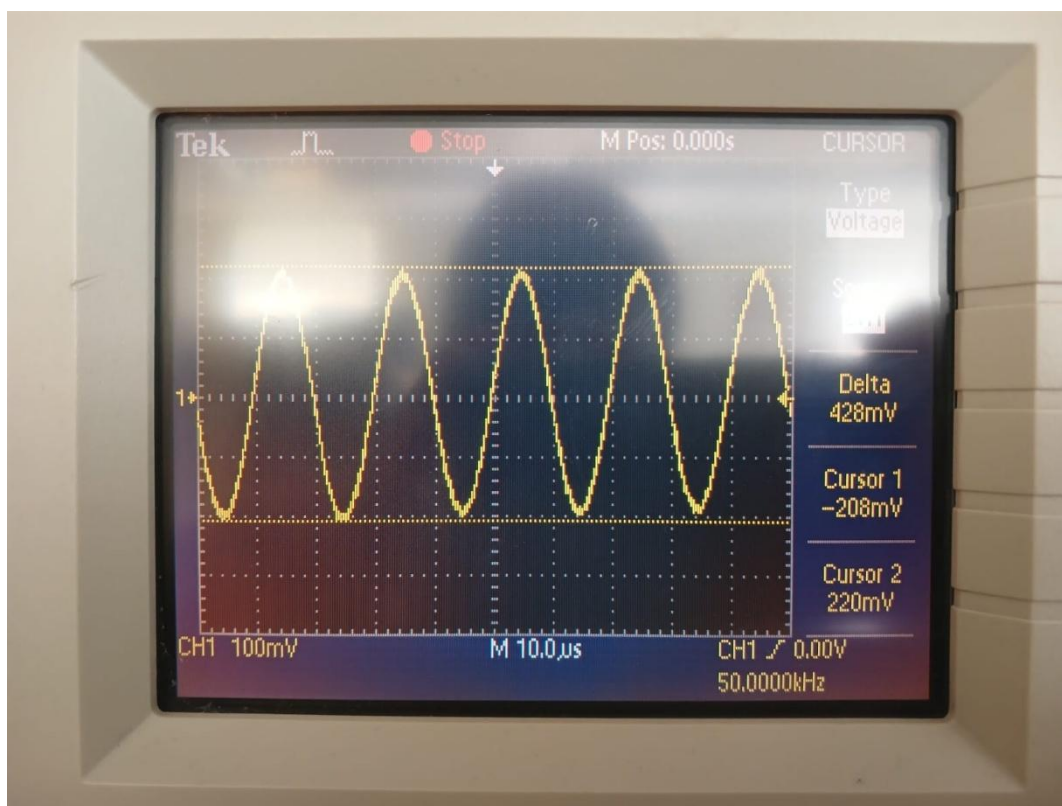


Figure 2.9: Output Voltage of the modified C-Type Circuit at f_3

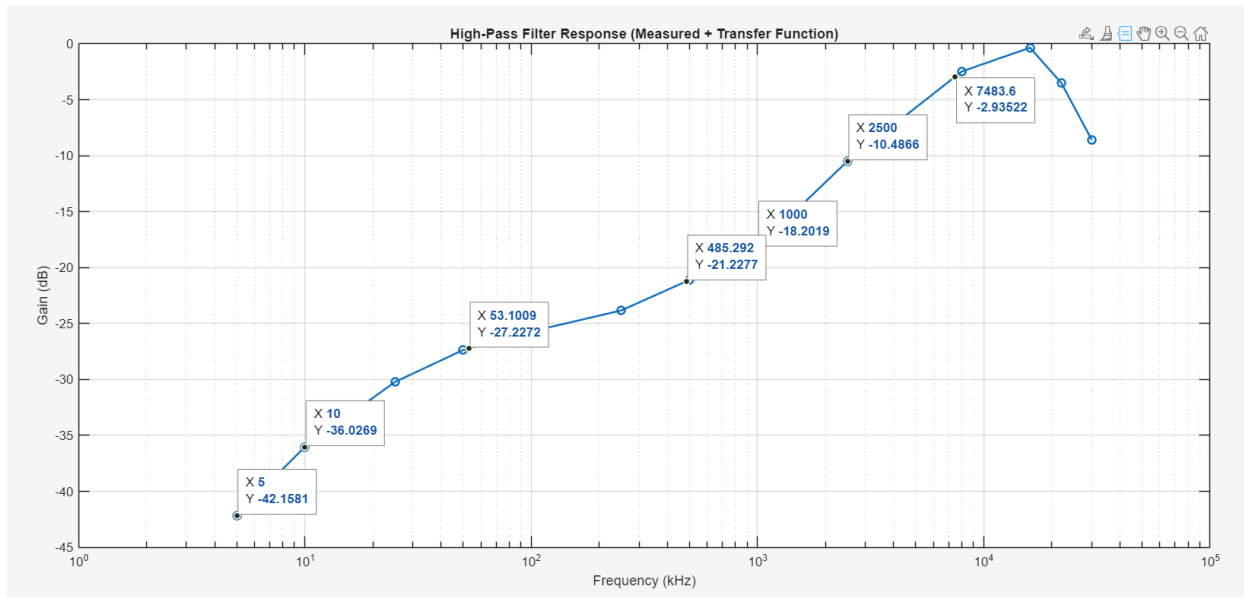


Figure 2.10: Semi-log plot of our data in Table 2 (Code in Appendix)

After these data entries were recorded, they were analyzed. Since, per the design instructions, we did not want to move the corner frequencies f_1 and f_2 , they were kept the same as the values found in the first circuit. As for f_3 , which was assigned to be 50 kHz, it was determined from the LTSpice plot as the point 6 decibels under f_1 since they are both corner frequencies of the same plateau. This was seen to be 50.3501 kHz from the LTSpice plot, yielding an error of 0.7002%. Additionally, from the plot, it was observed that the gains at corner frequencies f_1 and f_2 had slightly fluctuated. The gains were found to be -21.1930 dB and -2.7419, yielding an error of 2.6285% and 8.6125% for f_1 and f_2 respectively. These shifts could be caused by non-exact component values and the parasitic capacitance shifting the overall frequencies. After these calculations, we had gathered a strong basis to compare our measured data to.

From the MATLAB plot, the gains at the corner frequencies were determined to be -21.2277 dB and -2.9352 dB for f_1 and f_2 respectively, yielding an error of 0.1637% and 7.0499%. The low error rate of f_1 could be caused by human measuring errors, but the comparatively higher error for f_2 can be caused by the corner frequency being above 6 MHz, which is the rating of our oscilloscope probe. Additionally, since we have defined f_3 to be the frequency at which the gain is 6 decibels less than f_1 , from the plot, f_3 was found to be 53.1009 kHz, resulting in an error of 5.4633%. This could be caused by the capacitors' values not being exactly equal to the ideal values and measurement errors.

On top of our corner frequencies, the slopes of the graph were also analyzed. By looking at the frequency pairs of $0.1f_3$ and $0.2f_3$; $2f_1$ and $5f_1$, and the respective gains between them, slopes of 6.131 dB/octave and 5.8362 dB/octave were observed. Compared to the ideal slope of 6 dB/octave, respective errors of 2.1833% and 2.7300% were gathered. Due to the small error values, which can be the result of measurement errors or non-exact component values, we have determined the slope between low frequencies and f_3 ; frequencies between f_1 and f_2 to be approximately 6 dB/octave, which matches our theory. Finally, the overall shape of the MATLAB plot was very similar to the LTSpice simulation, with the exception of the last three data entries, which are high frequencies. This can be caused by the oscilloscope probes being rated for measurements until 6 MHz, and the data points exceeding that value.

Conclusion:

The purpose of this lab assignment was to design, analyze and simulate the frequency response of a C-type RL filter and its extended form with an added RC stage. First, the given design constraints and corner frequencies were examined, and appropriate component values were derived analytically. Using these values, the transfer function of the initial circuit was obtained and its response was plotted in MATLAB, where the expected gains at corner frequencies and asymptotic regions were observed. The design was then implemented in LTSpice to verify the analytical results, and the simulated responses showed close agreement, with small errors attributed to component approximations and numerical limitations. After completing the base design, an RC stage was added to introduce a third corner frequency, and a new transfer function was derived accordingly. The updated circuit was analyzed both analytically and through LTSpice simulations, where the desired frequency response, slopes, and corner frequencies were successfully achieved. Minor discrepancies were observed in gain and slope values, but these were within acceptable limits. From these results, we have gathered a strong basis for comparing our experimental results to the ideal cases for first and second order RL and RC filters.

After the preliminary section, the derived designs were implemented in real-life using the derived components and power was supplied to the circuit. For the first part, the frequencies that were assigned were supplied and the output voltage was measured. The gathered data was plotted using MATLAB and compared to an updated version of the LTSpice circuit. Some small errors were measured, which were attributed to non-exact component values and the ineffectiveness of our measuring equipment at high frequencies. The circuit was then modified as per the second part's design and measurements were conducted to see the response of the circuit over a wide interval. The same procedure was followed as the first part and similar error values were seen. Through this laboratory assignment, filter design steps were more deeply understood. Additionally, hardware implementation and circuit design skills were gained and improved.

References:

- R. E. Thomas, A. J. Rosa, and G. J. Toussaint, *The Analysis and Design of Linear Circuits*. Hoboken, NJ, USA: Wiley.
- H. Koymen and A. Atalar, *Analog Electronics*, ver. 4.2. Ankara, Turkey: Bilkent University, 2015.

Appendix:

Code for C-Type Filter (Figure B):

$R5 = 697.8983;$

$R6 = 47.7637;$

$L = 15e-6;$

$f = \text{logspace}(3,9,2500);$

$w = 2*\pi*f;$

$H = (R6 + 1j*w*L) ./ (R5 + R6 + 1j*w*L);$

```
semilogx(f, 20*log10(abs(H)));  
grid on;  
xlabel('Frequency (Hz)');  
ylabel('Gain (dB)');  
title('C-Type Filter Response');
```

Code for C-Type Data MATLAB Plot (Figure 1.9):

```
f = [49.86855, 99.7371, 199.4742, 398.9484, 498.6855, 598.4226, 800.01885, 997.371, 1600.0377,  
3200.0754, 6400.1508, 8000.1885, 9600.2262, 16000.377];  
Vout = [0.624, 0.63, 0.664, 0.792, 0.888, 1.06, 1.42, 1.80, 2.82, 5.20, 6.52, 7.44, 8.90, 7.42];  
Vin = 10;  
gain_dB = 20 * log10(Vout / Vin);  
figure;  
semilogx(f, gain_dB, 'o-', 'LineWidth', 1.5);  
grid on;  
hold on;  
xlabel('Frequency (kHz)');  
ylabel('Gain (dB)');  
title('High-Pass Filter Response (Measured + Transfer Function)');
```

Code for Modified C-Type Data MATLAB Plot (Figure 2.10):

```
f = [5, 10, 25, 50, 100, 250, 500, 750, 1000, 2500, 8000, 16000, 22000, 30000];  
Vout = [0.078, 0.158, 0.308, 0.428, 0.518, 0.643, 0.880, 1.06, 1.23, 3.00, 7.52, 9.60, 6.68, 3.72];  
Vin = 10;  
gain_dB = 20 * log10(Vout / Vin);  
figure;  
semilogx(f, gain_dB, 'o-', 'LineWidth', 1.5);  
grid on;  
hold on;  
xlabel('Frequency (kHz)');  
ylabel('Gain (dB)');  
title('High-Pass Filter Response (Measured + Transfer Function)');
```